$$= y^2(1-2xy+y^2)^{-3/2}$$

$$= y^2\left[(1-2xy+y^2)^{-1/2}\right]^3$$

$$\left[\because (1-2xy+y^2)^{-1/2} = u\right]$$

$$= y^2u^3$$

L.H.S. = R.H.S.

Hence proved.

10. If $z = \tan^{-1}(y/x)$ prove that

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0.$$

sol^n:-

$$z = \tan^{-1}(y/x)$$

diff. partially w.r.t. 'y'

$$\frac{\partial z}{\partial y} = \frac{\partial(\tan^{-1}(y/x))}{\partial y}$$

$$= \frac{1}{1+\left(\frac{y}{x}\right)^2}\frac{\partial(y/x)}{\partial y}$$